

# Graphing Basic Quadratic Equations

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A **quadratic relationship** is represented by an equation whose highest exponent of the variable is 2.

# Parabola

The graph of a quadratic equation is called a **parabola**.

A parabola is a smooth, U-shaped curve.

Vertex of  $y = ax^2$

The graph of

$$y = ax^2$$

has its **vertex** at  $(0, 0)$ .

# Direction of Opening

If  $a > 0$ , the parabola **opens upward**.

If  $a < 0$ , the parabola **opens downward**.

# Vertical Shift

In

$$y = ax^2 + b,$$

the graph is shifted **vertically** by  $b$  units.

The vertex moves to  $(0, b)$ .

# Horizontal Shift

In

$$y = a(x - h)^2,$$

the graph is shifted **horizontally**; its vertex is at  $(h, 0)$ .

# Width of the Parabola

The value of  $|a|$  affects the **width** of the parabola.

Larger values make it **narrower**, while smaller positive values make it **wider**.

# Graphing with a Table of Values

To graph a quadratic equation:

1. Make a **table of values**.
2. Plot the ordered pairs.
3. Draw a smooth curve.

# Axis of Symmetry

The **axis of symmetry** passes through the vertex of the parabola.

It is the vertical line that divides the parabola into two mirror-image halves.

# Examples

# Example 1

Make a table of values for  $y = x^2$  using  $x = -2, -1, 0, 1, 2$ .

$$x = -2 \rightarrow y = 4$$

$$x = -1 \rightarrow y = 1$$

$$x = 0 \rightarrow y = 0$$

$$x = 1 \rightarrow y = 1$$

$$x = 2 \rightarrow y = 4$$

## Example 2

Identify the vertex and direction of opening of  $y = -3x^2 + 2$ .

**Answer:** Vertex is  $(0, 2)$ . Since  $a = -3 < 0$ , the parabola opens **downward**.

## Example 3

Identify the vertex of  $y = (x - 2)^2$ .

**Answer:** Comparing to  $y = a(x - h)^2$ ,  $h = 2$ ,  
so the vertex is  $(2, 0)$ .

# Practice Exercises

# Practice 1

Complete a table of values and graph  $y = x^2$  using  $x = -3, -2, -1, 0, 1, 2, 3$ .

## Practice 2

Complete a table of values and graph  $y = -2x^2$  using  $x = -2, -1, 0, 1, 2$ .

## Practice 3

Complete a table of values and graph  $y = \frac{1}{2}x^2$   
using  $x = -4, -2, 0, 2, 4$ .

## Practice 4

Complete a table of values and graph  $y = x^2 + 4$ . Identify the vertex and state how the graph differs from  $y = x^2$ .

## Practice 5

Complete a table of values and graph  $y = -x^2 - 2$ . Identify the vertex and describe the direction of opening.

## Practice 6

Complete a table of values and graph  $y = (x - 3)^2$ . Identify the vertex and axis of symmetry.

## Practice 7

Complete a table of values and graph  $y = -2(x + 1)^2$ . Identify the vertex and direction of opening.

# Activities

# Activity 1

Complete a table of values and graph  $y = 2x^2$  using  $x = -3, -2, -1, 0, 1, 2, 3$ .

## Activity 2

Complete a table of values and graph  $y = -x^2$  using  $x = -2, -1, 0, 1, 2$ .

# Activity 3

Complete a table of values and graph  $y = \frac{3}{4}x^2$   
using  $x = -4, -2, 0, 2, 4$ .

# Activity 4

Complete a table of values and graph  $y = x^2 - 3$ . Identify the vertex and state how the graph differs from  $y = x^2$ .

# Activity 5

Complete a table of values and graph  $y = -x^2 + 5$ . Identify the vertex and describe the direction of opening.

# Activity 6

Complete a table of values and graph  $y = (x + 2)^2$ . Identify the vertex and axis of symmetry.

Thank you!